Q1) SaiTeja

a) Truth table → 2.5

+ tautology → 0.5

No truth table → 0

b) truth table → 1

tautology → 0.5

modus ponens → 0.5

No truth table → 0

c) CNF →

$(\neg P \vee Q \vee R) \wedge (P \vee \neg Q) \wedge (P \vee \neg R) \wedge (S \vee T)$

↳ with steps → 2

DNF

$(PQS) \vee (PQT) \vee (PRS) \vee (PRT) \vee$

$(\neg P \neg Q \neg R S) \vee (\neg P \neg Q \neg R T)$

with steps → 2

Q2abc: A

a) X → original bi

Y → read bit

$P(X=1|Y=1) = P$

$P(Y=1) = P(Y=1|X=1)P(X$

$P(Y=1|X=1) = \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} =$

$P(Y=1) = \frac{P}{2} + \frac{1}{4}$ →

b) Independent given M

$P(M|SH) = \frac{P(SH|M)}{P(SH)}$

c) i) $P(H=1|T=1)$

ii) Accuracy = 0.99 0.5

Precision = 0.5 1

Recall = 0.99 1

Q2 1) Path

Dist Calc

Diag → 0

KNN label + pts

Q2abc: Anurag

a) X → original bit
Y → read bit

$$P(X=1 \mid Y=1) = \frac{P(Y=1 \mid X=1)P(X=1)}{P(Y=1)} \quad 0.5$$

$$P(Y=1) = P(Y=1 \mid X=1)P(X=1) + P(Y=1 \mid X=0)P(X=0) \quad 0.5$$

$$P(Y=1 \mid X=1) = \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{3}{4} \;\; 0.5 \quad \Big| \quad P(Y=1 \mid X=0) = \frac{1}{4} \;\; 0.5$$

$$P(Y=1) = \frac{p}{2} + \frac{1}{4} \rightarrow 0.5 \quad \Big| \quad \text{Ans} \rightarrow \frac{3p}{2p+1}$$

Wrong ans → −0.5

b) Independent given M → $P(SH \mid M) = P(S \mid M)\,P(H \mid M)$

$$P(M \mid SH) = \frac{P(SH \mid M)\,P(M)}{P(SH)} \;\; 1 \;=\; \frac{P(S \mid M)P(H \mid M)P(M)}{P(SH)} \;\; 1 \;\; 1$$

c) i) $P(H=1 \mid T=1) = \dfrac{P(T=1 \mid H=1)\,P(H=1)}{P(T=1)}$ 1

ii) Accuracy = 0.99 0.5 $\quad = \dfrac{0.99 \times 0.01}{2(0.99 \times 0.01)}$ 1 $= \dfrac{1}{2}$ Wrong ans ↓ −0.5

Precision = 0.5 1

Recall = 0.99 1

Q2 d) Parth

Dist Calcs → 1

Diag → 1

KNN labels + pts → 1

5

0.5

0

0.5

→ 0.5

$(P \vee \neg R) \wedge (S \vee T)$

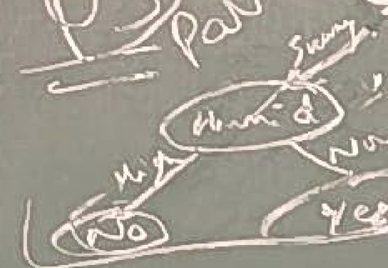
→ 2

$\vee (P R \neg T) \vee$

$P \neg Q \neg R \neg T)$

→ 2

Q3) Parth



1.5 each for
Outlook,

Q5) 1 Correct a

Tejasvini

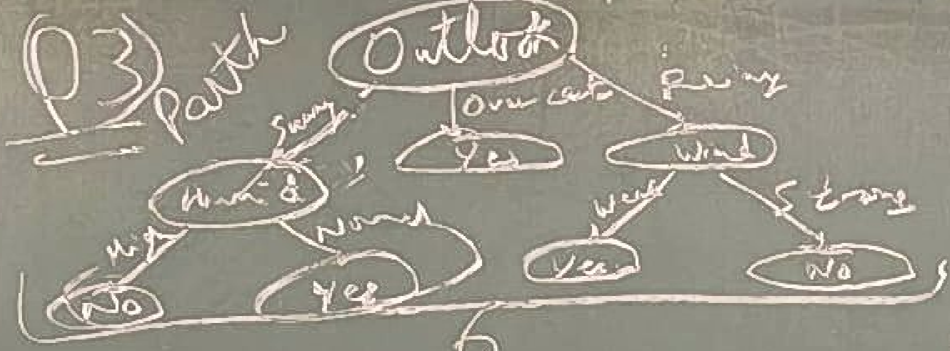
If weighted mean m
(10 out) Bias = −0

if combined bo
Bias = 0,

if Bernoulli me
for 10 out
for 20 out

For each correct
if method corr
if partially
if only form
+0.25 for f
if weighted me
no marks fo
if correct
• for combine
scaled by
for weighted
Go through
on reasonab

Q3) Part h



6

1.5 each for 4 of I's
Outlook, Temp, Humid, Wind

Q5) ① Correct ans: for 10 outcom = Bias = 0, Var = 0.0022, MSE = 0.0022
for 20 out Bias ≈ 0, Var = 0.00387 = MSE

Tejasvini

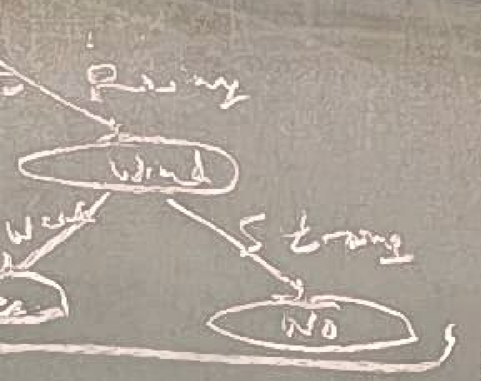
if weighted mean method used [final marks $\boxed{\times 0.9}$
(10 out) Bias = −0.1, var = 2.92 = MSE | 20 out → Bias = −0.15, Var = 2.93 = MSE

if combined both sets [final marks $\boxed{\times 1.6}$
Bias = 0, Var = 0.00147 = MSE
+1 +1 +1

if Bernoulli method used for var → full marks
for 10 out | Bias = 0, Var = 0.01366 = MSE
for 20 out | Bias ≈ 0, Var = 0.0067 = MSE

For each correct ans +1 for Bias +1 for Var +1 for MSE with brief steps shown.
if method correct, but calculation mistake − +0.75 each
if partially correct concept → +0.5
if only formulaes written → +0.25 each
+0.25 for finding probability of each 10 & 20 outcome
if weighted mean approach used, or calculation mistakes − no marks for probability calculations
if correct concept, but finally did not divide by 6 −0.25
• for combined approach calculated for 5 marks & scaled by 1.6
for weighted means approach → final ans ×0.9
Go through key before approaching TA. If unreasonable queries, you will lose marks

Q4 Mo...
a) Support...
Confide...
Lift...
b) Any a...
describe...
enables...
Mention...
c) E/2
Any va...
d) Option...
e) 1, 2, 3, ...
0.6 ma...
marks ...
not bee...
3 row...
by cas...



I 5

mid Wind

outcom = Bias = 0, Var = 0.0022,
MSE = 0.0022

20 out Bias ≈ 0, Var = 0.00387 = MSE

ed [final marks $\boxed{\times 0.9}$

= 2.92 = MSE] 20 out → Bias = -0.15
Var = 2.93 = MSE

s [final marks $\boxed{\times 1.6}$

.00147 = MSE

+1

sed for var → full marks

0, Var ≈ 0.01366 = MSE

≈ 0, Var = 0.0067 = MSE

+1 for Bias +1 for Var +1 for MSE with brief steps shown -

t calculation mistake - +0.75 each

ect concept → +0.5

es written → +0.25 each

g probability of each 10 & 20 outcome

pproach used, or calculation mistakes -

robability calculations

ept, but finally did not divide by 6 -0.25

pproach calculated for 3 marks &

.6

means approach → final ans X0.9

before approaching TA. If

u will lose marks

Q4 Monday

a) Support = 1/2 0.66 marks
Confidence = 3/4 0.66 marks
Lift = 9/8 0.66 marks

b) Any answer which describes that ε greedy enables exploration - 0.5 mark
Mentioning "exploration" - 0.5 mark

c) ε/2 - 0.5 mark
Any valid reasoning ! 0.5 mark

d) Option b - 1 mark

e) 1, 2, 3, 1, 2, 2, 2, 3, 1, 3
0.6 marks per round. No marks if round robin has not been performed in first 3 round. Otherwise marks given by case to case basis

Q6 A) 70   100

4 [2M Optimal [70 70 70 70] [100 100 100 100]
    2M ML [70 70 70 70] [100 100 100 100]

4 [2 B) Optimal [40.93, 48.39, 59.02, 70] [61.23, 72.52, 85.37, 100]
    2 ML [11.07, 15.39, 59.02, 70] [21.43, 27.18, 85.37, 100]

c) 1.5 + 1.5 for calc MSE for both parts.

2 Mthds of Solving: ① Assume $[u1^*, u2^*, u3^*, 70]$ & write eqN. → 4 Equation Solving

② Value iteration from [0, 0, 0, 70]

1/2 marks of each correct iteration (at Max for 2 iteration)

A) Optimal
- V1: [0 0 56 70] 1/2
- V2: [0 44.8 67.2 70] 1/2

ML
- V1: [0 0 56 70] 1/2
- V2: [0 5.6 67.2 70] 1/2

and so on.

Q6 ABC MANAS

Q7 : Krishiv

(a) T/F → 0.5

Explanation – 1.5

Answer (60) – 1

(b) T/F — 0.5

Risk neutral modelling – 1 (0.5 given if just explained about it)

For explaration & computation – 1.5

(c) T/F — 0.5

Risk averse modelling – 1

Exp^n and comp^n – 1.5

(d) (0.52, 0.48) – 1 mark

Final Ans – 1 mark

0.5 – Utility fn

0.5 – Won't bet

Q8) Jai

a) Obj w/o R — 1

Explain — 1

Obj with R — 1

Explain — 1

b) FE — modify/improve features — 1

Scaling — Scale to fixed range — 0.5

Ext any standard technique — 0.5

Binning — Continuous fts. to discreet bins — 1

Q6d) Jai

Dim = 4 × 13 — 0.5

S_4 reach col. — 0.5

0.25 / col.

Extra col. — −0.5

Q10) Tathya

Alt 1: Manhattan

a) $h(n) = 2(4-x) + 2($

Alt 2: h_{SLD}

Alt 3: $h(n) = h($true

or any other heuristic fun

(x, y) or (r, c) to $(x_g,$

For showing

$h(n) \leq h^*(n)$

For showing

Consistency —

All 4 cases

b) Expansion

Alt 1: M

Alt 2: h_{S}

Alt 3: $h(n$

1 mark for (

2 marks for s

c) GBFS has

Correct $h(n)$

d) GBFS has

more nodes ex

doesn't guaran

its greedy algo

the best algo w

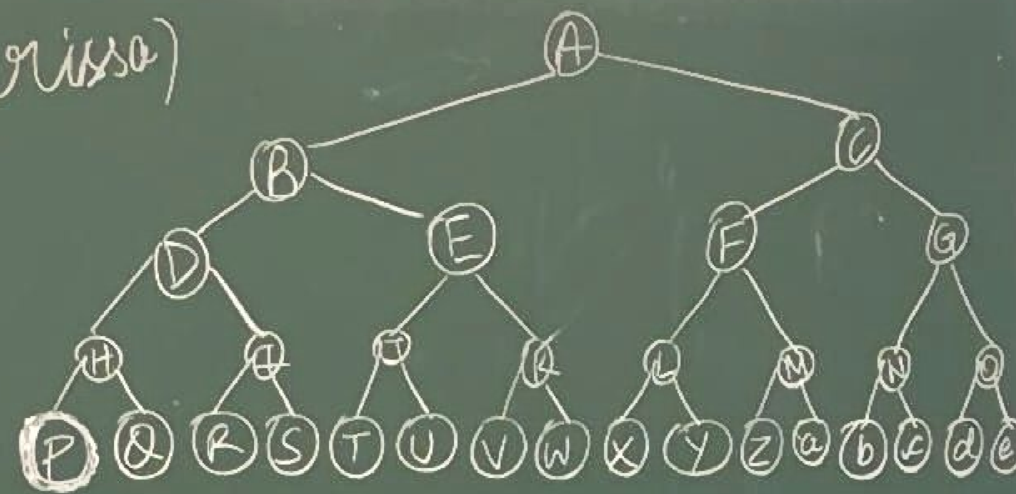
ya.
on
) + 2(c−1)
— ①
h(truecost)
ristic from
to (x_g, y_g)
$h^*(n)$ — ①

cy — ①
ases
sion
1: Manhattan has 13 nodes
2: h_{SLD} has 10−12 nodes
3: $h^*(n)$ has 6 nodes
k for correct no of nodes
s for steps correct g(n)&h(n) values

FS has around 6 nodes — ①
t h(n) values in steps — ①

FS has 6 while A^* has
nodes expansion but GBFS
t guarantee optimal sol/n
dy algo & A^* is optimally efficient → 0.5
t algo among all other algo that
... cuz its consistent &

9) (Larissa)



DFS stack (1 mark): A → CB → CED → CEIH → CEIQP
⇒ smmDFS = $\boxed{5 \text{ (1 mark)}}$

BFS queue (1 mark): A → CB → EDC → GFED → ... → edcbaZYXWVUTSRQP
⇒ smmBFS = $\boxed{16 \text{ (1mark)}}$

smmBFS − smmDFS = 16 − 5 = $\boxed{11 \text{ (0.5 marks)}}$

Reasoning (1.5 marks):

The chosen goal node gives maximum value since smm BFS depends on width of level of node and smm DFS depends on depth.

smmBFS: exponential increase on going down
smmDFS: linear increase on going down

Goal node: any in last level

y W/OR — 1

plain — 1

with R — 1

plain — 1

— Modify/improve features — 1

ing — Scale to fixed range — 0.5

Ext any standard technique — 0.5

nning — Continuous pts. to discreet bins — 1

air

mm = 4 X 13 — 0.5

noop rol. — 0.5

0.25/col.

Extra col. — -0.5

Q10) Tathya

a) Alt 1: Manhattan $h(n) = 2(4-r) + 2(c-1)$

Alt 2: h_{SLD}

Alt 3: $h(n) = h(\text{true cost})$ — 1

or any other heuristic from (x,y) or (r,c) to (x_g, y_g)

For showing $h(n) \leq h^*(n)$ — 1

For showing Consistency — 1
All 4 cases

b) Expansion
Alt 1: Manhattan has 13 nodes
Alt 2: h_{SLD} has 10-12 nodes
Alt 3: $h^*(n)$ has 6 nodes
1 mark for correct no of nodes
2 marks for steps correct $g(n)$ & $h(n)$ values

c) GBFS has around 6 nodes — 1
Correct $h(n)$ values in steps — 1

d) GBFS has 6 while A^* has more nodes expansion but GBFS doesn't guarantee optimal soln → 0.5
its greedy algo & A^* is optimally efficient the best algo among all other algo that can find optimal soln cuz its consistent & admissible → 0.5

9) (Larissa)

DFS stack (1 mark): A → ... ⇒ sm...

BFS queue (1 mark): A → ...

⇒ smmBFS = 16

smmBFS - smmD...

Reasoning (1.5...

The chosen goal ... since smm BFS ... node and smm ...

smmBFS: expone...
smmDFS: linear...

Goal node ...